



Generic Windows Service, Version 1.0

User's Guide

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1 Introduction

For a program to run as a Windows service, the program must be written specifically as a Windows Service. This requires the program to be written in C to access the Windows API so Windows recognizes the program as a service. The Generic Windows Service Controller is a native executable that adheres to the Windows SDK rules and guidelines for a service and acts as a facade to the Windows API.

The Platform Integration Toolkit Generic Windows Service removes the need for the developers to write C code to run their program as a service. The Generic Windows Service runs any executable as a Windows service.

1.1 Product overview

To satisfy the platform integration requirements, server products need to be able to run as a service. This ensures the server runs at system startup. This component enables developers to run an executable as a service without the need to write C code to access the Windows API. All the developer needs to do is configure the Generic Windows Service so it knows which commands to run to start and stop the service, and register the service to the Windows Service Controller.

The Generic Windows Service component consists of two files, `AZY_GenericWindowsService.exe` and `AZY_GenericWindowsService.ini`. The `.ini` file contains the configuration information for the service. This file includes information about the name of the service, the working directory, the start command, and the stop command.

1.2 Features and benefits

The Generic Windows Service enables products to run any executable as a Windows Service. This eliminates the need to write C code to access the Windows API to run the program as a service.

1.3 Intended audience

This document is intended for the IBM developer who needs to run a program as a Windows service and does not need to write a Windows service specific program written in C.

1.4 Requirements

1.4.1 Hardware requirements

The Generic Windows Service component has no specific hardware requirements.

1.4.2 Software requirements

The Generic Windows Service component runs on the following operating systems:

- Microsoft Windows
 - Windows XP Professional SP2
 - Windows Server 2003 SP1
 - Windows Server 2003 R2
 - Windows Server 2003 SP1 Enterprise Edition
 - Windows Server 2003 R2 Enterprise Edition
 - Windows Server 2003 Server

2 Using the Generic Windows Service component

To install a service using the Generic Windows Service component, perform the following steps:

1. Place the `AZY_GenericWindowsService.exe` and `AZY_GenericWindowsService.ini` files in a location of your choosing. You can rename these files if you want.
2. Edit the `AZY_GenericWindowsService.ini` file to specify the name of the service, the working directory, the start command, and the stop command. Contents of an example .ini file:

```
[Service]
ServiceName=IBMProductServiceName
WorkingDirectory="C:\Program Files\IBM\ProductName"
StartCommand="""C:\Program Files\IBM\ProductName\jre\bin\javaw.exe" -Xrs
-jar product.jar -start"
StopCommand="""C:\Program Files\IBM\ProductName\jre\bin\javaw.exe" -Xrs
-jar product.jar -stop"
```

3. Register the Generic Windows Service with the Windows Service Controller by either:
 - a. Using the `sc.exe`¹ command. (Example: `sc.exe create "IBMProductServiceName" binPath= "C"\Program Files\IBM\ProductName\service\AZY_GenericWindowsService.exe" DisplayName= "IBM Product Service Display Name"`)
 - b. Using the Macrovision InstallShield's bean "Windows Service"
4. (Uninstall) Unregister the Generic Windows Service with the Windows Service Controller by either:
 - a. Using the `sc.exe` command. (Example: `sc.exe delete "IBMProductServiceName"`)
 - b. Using the Macrovision InstallShield's bean "Windows Service"

To uninstall the service, perform the following steps:

1. Stop the Generic Windows Service using the following command:
`sc.exe stop "IBMProductServiceName"`
2. Unregister the Generic Windows Service with the Windows Service Controller by using the following command:
`sc.exe delete "IBMProductServiceName"`

¹ The `sc.exe` command is a Windows system executable that can be run from a Windows Command prompt.

3 Notes and Limitations

When the Windows Service Controller tells the Generic Windows Service to start, the Generic Windows Service executes the start command and indicates to the Windows Service Controller the service is started. During this time, the Generic Windows Service does not expect the process to return. If the process is returned back to the Generic Windows Service, it will think the service has stopped and will report to the Windows Controller that the service has stopped. If the behavior of the application is to return the process while it is executing, the end result is the service will look like it is stopped in the Window's Services view when in fact the application is still running.

The Generic Windows Service component suppresses any windows that pop up during the execution of the start and stop commands. You should not execute anything that prompts the users for input as it hangs the process.

4 Reference

<http://rchgsa.ibm.com/~midmarkt/express/>

This is the Express Design and Architecture Web site. This Web site contains information on Platform Integration Toolkit as well as the Express architecture.

<http://rchgsa.ibm.com/~midmarkt/components/>

This is the Midmarket component site containing information on various Express enablement components.

<https://cs.opensource.ibm.com/projects/platinttoolkit/>

This is the IBM Community Source Platform Integration Toolkit project web site. This Web site contains the latest releases and information for the Platform Integration Toolkit.